

# **StockSmart**

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## What is Inventory Management?

- Tracking goods from purchase through to sale
- Covers stock ordering, storage & replenishment
- Core for shops, supermarkets & warehouses
- Directly affects revenue and customer satisfaction

### WHY IT MATTERS

**43%**

of SMEs face monthly stock-outs

**\$1.1T**

lost globally from poor stock control

**30%**

SME losses from expired/wasted goods

**2 in 3**

SMEs still use manual tracking methods

## Who Are SMEs and What Do They Face?

- **Small Shops**

Limited staff; rely on memory or paper logs

- **Warehouses**

Large volumes; manual tracking causes costly errors

- **Supermarkets**

High product variety; expiry tracking is critical

- **Wholesalers**

Seasonal demand spikes lead to over/under-stocking

### **Why StockSmart is the Answer:**

- Centralized system – one dashboard for all stock
- Real-time updates – no delays or manual entry
- Affordable and easy to learn
- Designed specifically for SME workflows

### Key Challenges Faced by SMEs Today

- **Stock Shortages**  
Business are unable to meet customer demand, resulting in revenue loss
- **Overstocking**  
Excess inventory increases storage costs and blocks capital
- **Expired Products**  
Lack of tracking leads to wastage and financial loss
- **No Real-Time Visibility**  
Stock levels cannot be monitored accurately
- **Manual Record Errors**  
Paper and spreadsheets increase chances of human mistakes



#### **Paper 1:Web-Based Inventory Platform for SMEs (Source: Agboola, Malgwi, Mahmud & Oguntoye (2022))**

- Helps replacing spreadsheets with digital records to reduce errors
- Includes product tracking, sales management, and user management modules
- Improved inventory accuracy and reduced operational cost for SMEs

Limitations:

- It may not work the same in all industries.
- Less flexible for changing business.
- It does not have automatic expiry alerts.

#### **Paper 2: ERP Implementation for Supply Chain Management (Source: Qureshi (2022))**

- Identifies the key factors for successfully implementing Enterprise Resource Planning, in supply chains and ranks them based on importance.
- It uses ISM and MICMAC methods to show how the key factors are connected.
- ERP systems help businesses grow and make supply chains better.

Limitations:

- ERP systems are extremely costly and complex which are impractical for small enterprises
- ERP systems take a long time to set up and require skilled IT staff
- Study focuses on only large organizations

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### **Paper 3: IoT-Based Inventory System (Source: Ghulam Haider & Edward Lucas (2024))**

- IoT-based automated inventory tracking system
- Uses sensors for real-time stock monitoring
- Reduces manual inventory effort

Limitations:

- Requires IoT hardware (sensors/RFID)
- High setup and maintenance cost
- Technical complexity for SMEs

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### **Paper 4: AI-Based Inventory System (Source: Abdul Majid K & Kavyashree G.J (2025))**

- Machine learning-based inventory management system
- Uses forecasting (Prophet), pattern mining (Apriori), prediction (Random Forest)
- Achieves high accuracy in demand prediction (~94%)

Limitations:

- Requires large historical datasets
- High computational complexity
- Does not address expiry-date or batch-level tracking

## 4.OBJECTIVES

Develop a simple and centralized inventory management system for SMEs

Provide automated alerts for low stock and product expiry

Enable real-time inventory tracking, updating, and transaction recording

Improve accuracy and operational efficiency in inventory management

Reduce losses from overstocking, stock shortages, and expired products

Lower overhead costs for businesses through automation



### **STEP 1 — Requirement Analysis**

- Analyze common inventory challenges faced by SMEs
- Identify key functional requirements for the system

### **STEP 2 — Literature Review & Gap Analysis**

- Study 4 existing inventory management systems
- Identify features missing from current solutions

### **STEP 3 — System Design**

- Design database schema, user flow, and UI wireframes
- Define core modules: inventory, alerts, dashboard, reports

### **STEP 4 — Feasibility Analysis**

- Evaluate technical, operational, and economic feasibility

### **STEP 5 — Development & Testing**

- Build and test the system for performance and accuracy
- Validate with sample data from real SME scenarios

**The feasibility analysis has been divided into three distinct sections:**

**Technical:**

- Tried and tested tech stack with all necessary documentation
- Knowhow of much of the tech stack
- Simple implementation for SMEs
- Mature libraries and APIs for sub-programs

**Operational:**

- Minimal costs associated with operation
- Real time inventory-tracking keeps costs low
- Integration with E-paper based price labels will reduce operational costs for clients

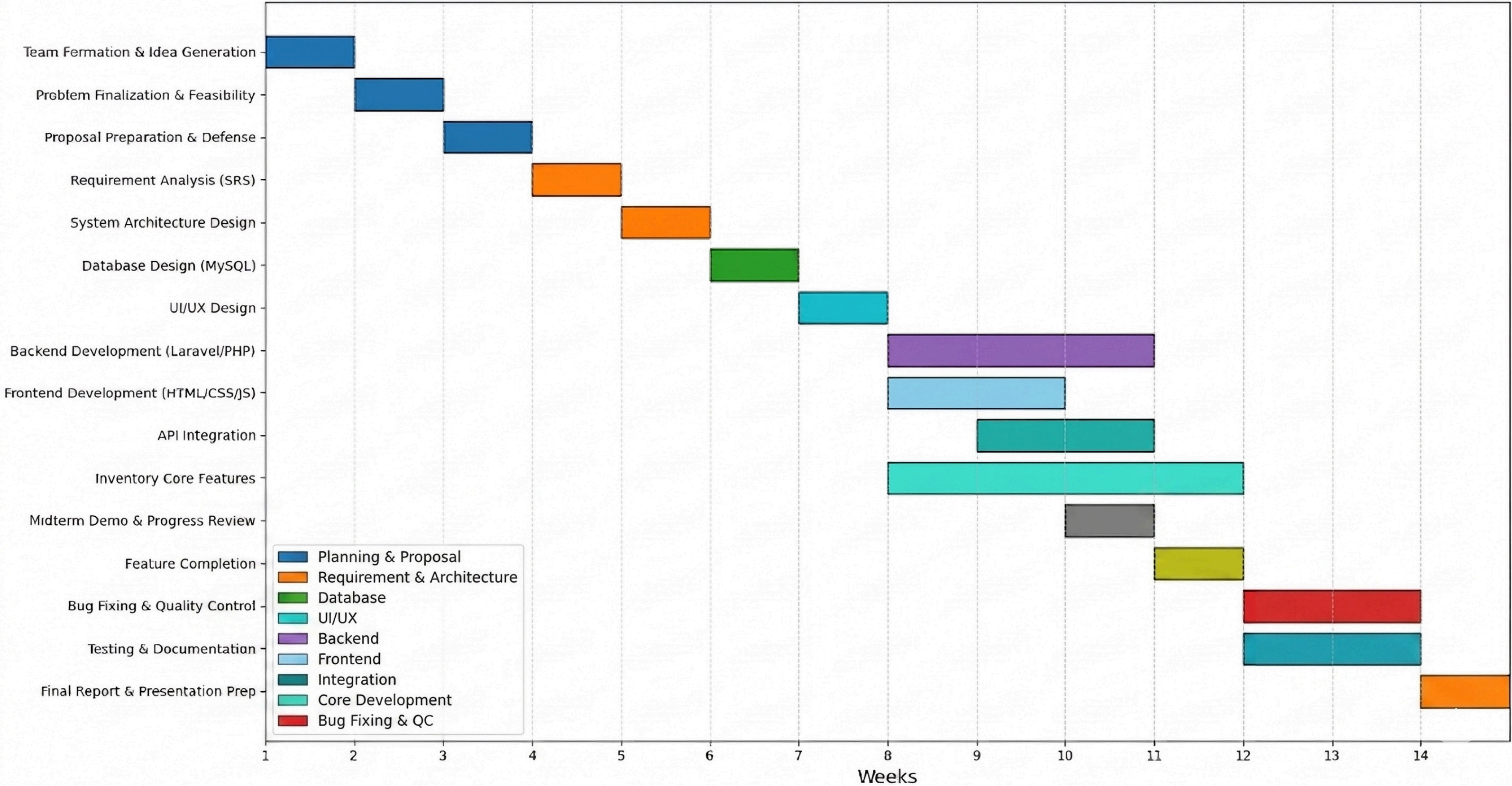
**Economic:**

- Self-Hosting will eliminate need for VPS, public ip, and/or domain, keeping costs low
- “maintenance” will be carried out till bugs are fixed
- Open-Source software results in no costs for capable clients
- Free payment processors will keep overheads low

# 7.GANTT CHART



Inventory Management System - 14 Week Gantt Chart





## Selected the LAMP Tech Stack based on Simplicity and familiarity

### L: Linux

- A headless linux ISO or a docker image will be used to run the rest of the stack

### A: Apache

- We will be using an apache web server like XAMPP for development locally

### M: MySQL

- MySQL will be used for CRUD operations involving the products and their inventorying

### P: PHP

- Laravel, a PHP framework will be used for backend and integration with MySQL

# 8.COMPARISON



| <u>Feature</u>      | <u>AI-Based</u> | <u>IoT-Based</u> | <u>Web-Based</u> | <u>ERP</u> | <u>StockSmart</u> |
|---------------------|-----------------|------------------|------------------|------------|-------------------|
| Cost                | High            | Very High        | Medium           | Very High  | Low               |
| Complexity          | High            | High             | Medium           | Very high  | Low               |
| Requires Hardware   | No              | Yes              | No               | No         | No                |
| Requires Large Data | Yes             | No               | No               | No         | No                |
| Ease of Use         | Low             | Medium           | Medium           | Low        | High              |

- **Replaces manual inventory with a simple digital system**
- **Provides real-time stock tracking and updates**
- **Reduces losses with alerts for low stock and expiry**
- **More affordable and easier than complex systems**
- **Improves efficiency and supports business growth**

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